

**23. Answer: D**

$$\begin{aligned}\text{Potential difference across the } 6.0 \, \Omega \text{ resistor} &= 6.0 - 0.50 \times (1.0 + 9.0) \\ &= 1.0 \, \text{V}\end{aligned}$$

$$\text{Current passing through the } 6.0 \, \Omega \text{ resistor} = \frac{1.0}{6.0} \, \text{A}$$

$$\text{Thus unknown current } I = 0.5 - \frac{1.0}{6.0} = 0.33 \, \text{A}$$

$$\text{Unknown resistance } R = \frac{1.0}{0.33} = 3.0 \, \Omega$$